UART Gesture-Controlled Modes Firmware

This project implements a firmware that receives gestures via UART, processes them, and executes commands based on the active mode. The supported modes are:

1. Palm PCB Control
2. Mouse Control (currently not Implemented)
3. Gamepad Control (currently not Implemented)
4. VR Control (currently not Implemented)

Project Features

- Handles four different control modes using an interrupt-driven mode switch mechanism.
- Supports debounced mode switching with an external button.
- Processes six distinct gesture commands (1-6).
- Executes unique actions per mode:
  - Palm commands (e.g., G1, G2).

Setup

1. Hardware Connections

- UART Connections:
  - UART0 (MANAR_UART_RX RX2 D16 and MANAR_UART_TX TX2 D17): Connect this to your UART tool or device for sending and receiving commands.
  - Make sure to common the ground connection.
- Button for Mode Switching:
  - Connect a momentary push button to the GPIO pin defined for MODE_SWITCH_PIN in the firmware (default: GPIO_NUM_23).
- LED (Optional):
  - Connect an LED to GPIO_NUM_2 (ESP32 BUILTIN LED) defined in the firmware to visualize mode switching (e.g., blinking interval 200ms).

2. Hardware Requirements

- ESP32 Development Board.
- Push button for mode switching.
- LED (optional, for visual feedback).

- UART communication device (e.g., Palm PCB).

How to Test

1. Flash the Firmware

To test the project, you can flash the ESP32 using the provided binary files. The files should be placed at the following addresses:
Flash Commands
- Bootloader: bootloader.bin → Flash at address 0x1000
- Firmware: palm_pcb_control.bin → Flash at address 0x10000
- Partition Table: partition-table.bin → Flash at address 0x8000

These binaries can be flashed using the following command (using esptool.py):

```bash
esptool.py --port /dev/ttyUSB0 write_flash 0x1000 bootloader.bin 0x10000
palm_pcb_control.bin 0x8000 partition-table.bin
```

- Power the ESP32 and monitor logs via UART.
- Use a UART terminal program (e.g., CoolTerm) to send commands to the ESP32.
- Set UART terminal program baudrate to 38400.

2. Send Gesture Commands

- Commands 1, 2, 3, 4, 5, and 6 represent gesture inputs.
- Send these commands over UART (configured with 38400 baud) to test different responses for the current mode.

3. Switch Modes

- Button GPIO 23
- Press the mode-switching button to change the mode.
- Interrupt on Negative Edge.
- Debounce Period 200ms.
- The mode order is:
  MODE_PALM -> MODE_MOUSE -> MODE_GAMEPAD -> MODE_VR -> MODE_PALM (loop)
- Debounce ensures that a single button press doesn't cause multiple mode switches.

Mode Details and Behavior

Palm PCB Control Mode

- Gesture commands (1-6) are mapped to G1 to G6.
- Each command is forwarded to another UART (MANAR_UART_TX).
  Example:
  - Input: 1
  - Output: G1 written to UART.

Mouse Control Mode

(currently not Implemented)

Gamepad Control Mode

(currently not Implemented)

VR Control Mode

(currently not Implemented)